

Adhesive Application Guide

Automotive Industry Applications

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Introduction

This guide provides best practices for applying TechBond adhesives in automotive manufacturing applications. Following these guidelines ensures optimal bond performance and production efficiency.

Product Selection Guide

By Application

Application	Recommended Product	Key Feature
Body-in-White	BondMax 500	High strength, paintable
Closure Bonding	BondMax 300	Flexibility, fatigue resistance
Hem Flange	BondMax 300	Gap filling, corrosion resistance
Roof Bonding	BondMax 500	Structural strength
Battery Assembly	SecureSeal EV-200	Thermal management
Interior Trim	FlexiGrip 200	Low odor, fast cure

By Substrate

Substrates	Product Options
Steel to Steel	BondMax 300, 500, 700
Aluminum to Aluminum	BondMax 500, 700
Steel to Aluminum	BondMax 500
Composite to Metal	BondMax 700, FlexiGrip 300
Plastic to Metal	FlexiGrip 200, 300

Surface Preparation

Metal Substrates

Step 1: Cleaning

- Remove oils, greases, and cutting fluids
- Use approved solvent (isopropyl alcohol or MEK)
- Wipe with clean, lint-free cloth

Step 2: Degreasing (if required)

- Alkaline wash for heavily contaminated surfaces
- Rinse thoroughly with clean water
- Dry completely before adhesive application

Step 3: Surface Activation (optional)

- Abrade with 80-120 grit sandpaper for maximum adhesion
- Remove dust with dry air
- Apply within 4 hours of preparation

Coated Metals (E-coat, Primers)

- Ensure coating is fully cured
- Light abrasion may improve adhesion
- Test adhesion to coating first
- Consult TechBond for compatibility

Application Methods

Manual Application

Suitable for:

- Low-volume production
- Prototype builds
- Repair applications

Equipment:

- Cartridge guns (400ml or 600ml)
- Static mixers (provided with cartridges)
- Spatulas for spreading

Automated Application

Suitable for:

- High-volume production
- Consistent bead application

- Robotic assembly lines

Equipment Requirements:

- Meter-mix dispensing system
- Heated hoses (if required for viscosity)
- Robot-mounted applicator head
- Bead inspection system (recommended)

Process Parameters

Temperature Recommendations

Parameter	Range	Optimal
Substrate Temperature	15-35°C	20-25°C
Adhesive Temperature	20-30°C	23°C
Ambient Temperature	15-35°C	20-25°C
Cure Temperature	20-80°C	Room temp or 60°C

Application Guidelines

Parameter	BondMax 300	BondMax 500
Bead Width	6-12 mm	8-15 mm
Gap Fill	Up to 3 mm	Up to 5 mm
Open Time	30 min	45 min
Fixture Time	20 min @ 80°C	30 min @ 80°C

Quality Control

In-Process Checks

1. Bead Inspection

- Continuous bead without gaps
- Consistent width and height
- Proper positioning

2. Cure Verification

- Shore hardness measurement
- Visual inspection
- Non-destructive testing

Destructive Testing

- Peel tests (per ASTM D1876)
- Lap shear tests (per ASTM D1002)
- Test frequency: Per production batch

Troubleshooting

Issue	Possible Cause	Solution
Poor adhesion	Surface contamination	Improve cleaning process
Slow cure	Low temperature	Increase ambient/oven temp
Bubbles in bond	Air entrapment	Check mixing, reduce speed
Inconsistent bead	Equipment issue	Calibrate dispenser

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